

SQL Fundamentals

Select and Filtering

Question 1: Retrieve all columns from the employees table

Query: Select *
From Employees
Limit 5;

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	65 000	2019-03-22	Cape Town
3	Cathy	Ndlovu	Finance	65 000	2018-07-18	Durban
		Mokoena	Marketing	50 000	2021-11-03	Pretoria
		Maidoo	IT	72 000	2017-09-30	Johannesburg

2. Find all unique departments

Query:
Select distinct department
FROM Employees;

Department
IT
HR
Marketing
Finance

3. Retrieve first and last name ordered by salary descending

Query:
Select first_name, last_name, salary
FROM Employees
Order BY Salary DESC;

First_name	Last_name
Eva	Maidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. Retrieve the top 3 highest-paid employees
 Query: Select first_name, last_name, salary, id
 FROM Employees
 Limit 3 Order BY Salary DESC.
 Limit 3.

id	first_name	Last_name	Salary
5	Eva	Maidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

5. Find employee in the IT department
 Query: Select id, first_name, last_name, department
 FROM Employees
 Where Department = IT;

id	first_name	Last_name	Department
1	Alice	Green	IT
5	Eva	Maidoo	IT

6. Find employees in Finance with salary 76000

Query: Select id, first_name, last_name, department, salary
 FROM Employees
 Department = Finance
 Salary 76000;

10. Find employees in IT with salary > 65000 and city Johannesburg

Query: Select id, ^{first}name, last_name, Department, salary, city
FROM Employees
Where Department = IT,
Salary > 65000 AND City = Johannesburg

id	first_name	last_name	department	salary	City
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg

id	first-name	last-name	Department	Salary
3	Cathy	Zulu	Finance	65000

7. Find employees in HR or Marketing

Query: Select id, first-name, last-name, department
FROM Employees

Where department = HR or Department = Marketing

id	first-name	last-name	Department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. Find employee not in IT

Query: Select id, first-name, last-name, Department
FROM Employees

Where Department NOT IT

id	first-name	last-name	Department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9 Find employees in IT, HR, or Finance using IN.

Query: Select id, first-name, last-name, department
FROM Employees

Where department IN IT, HR OR Finance;

id	first-name	last-name	Department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Maidoo	IT